

D. Allan Drummond, Ph.D.

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ACADEMIC APPOINTMENTS

2006–2011 Bauer Fellow, FAS Center for Systems Biology, Harvard University
2011–2017 Assistant Professor, Department of Biochemistry and Molecular Biology, University of Chicago (primary)
2011–2019 Assistant Professor, Department of Human Genetics, The University of Chicago (secondary)
2017–2026 Associate Professor with tenure, Department of Biochemistry and Molecular Biology, The University of Chicago (primary)
2019–2026 Associate Professor, Department of Medicine, Section of Genetic Medicine, The University of Chicago (secondary)
2026– Professor, Departments of Biochemistry and Molecular Biology (primary), and Department of Medicine, Section of Genetic Medicine (secondary), The University of Chicago

Ph.D.-Granting Committee, Program, Institute, and Center Appointments

2015–2018 Fellow, Institute for Genomics and Systems Biology
2014– Committee on Genetics, Genomics & Systems Biology
2022– Member, Institute for Biophysical Dynamics
2025– Member, National Institute for Theory and Mathematics in Biology

ACADEMIC TRAINING

1991–1995 B.S.E., Mechanical and Aerospace Engineering, Princeton University, Princeton, NJ
2002–2006 Ph.D., Computation and Neural Systems, California Institute of Technology, Pasadena, CA

SCHOLARSHIP

(a) Peer-reviewed publications in the primary literature, exclusive of abstracts:

1. Glauninger*, H., Bard*, J.A.M., Wong Hickernell*, C.J., Velez, K., Airoidi, E.M., Li, W., Singer, R., Paul, S., Fei, J., Sosnick, T.R., Wallace+, E.W.J., **Drummond+, D.A.** (* co-first, + co-corresponding), "Transcriptome-wide mRNP clustering precedes stress granule formation and excludes new mRNAs," *Molecular Cell* 35(23):4393–4409 <https://doi.org/10.1016/j.molcel.2025.11.003> (2025).
2. Bard, J.A.M., **Drummond, D.A.**, "Chaperone regulation of biomolecular condensates," *Frontiers in Biophysics* 2, 1342506 (2024).
3. Chu, J., Ejaz, A., Lin, K.M., Joseph, M.R., Coraor, A.E., **Drummond, D.A.**, Squires, A.H. "Single-molecule fluorescence multiplexing by multi-parameter spectroscopic detection of nanostructured FRET labels," *Nature Nanotechnology* 19:1150–1157 (2024).

4. Chen, R., Glauninger, H., Kahan, D., Shangguan, J., Sachleben, J.R., Riback, J.A., **Drummond, D.A.**, Sosnick, T.R. "HDX-MS finds that partial unfolding with sequential domain activation controls condensation of a cellular stress marker," *Proc. Nat'l. Acad. Sci. USA* 121(13):e2321606121 (2024).
5. Keyport Kik, S., Christopher, D., Glauninger, H., Wong Hickernell, C., Bard, J.A.M., Lin, K.M., Squires, A.H. Ford, M., Sosnick, T.R., **Drummond, D.A.**, "An adaptive biomolecular condensation response is conserved across environmentally divergent species," *Nature Communications* 15:3127 (2024).
6. Triandafillou, C.G., Pan, R., Dinner, A.R., **Drummond, D.A.**, "Pervasive, conserved structure in highly charged protein regions," *PLOS Computational Biology* 19(10): e1011565 (2023).
7. Ali, A., Garde, R., Schaffer, O.C., Bard, J.A.M., Husain., K., Keyport Kik, S., Davis, K.A., Luengo-Woods, S., **Drummond, D.A.**, Squires, A.H., Pincus, D., "Preservation of orphan ribosomal proteins during stress in chaperone-stirred condensates," *Nature Cell Biology* 25:1691–1703 (2023).
8. Glauninger, H., Hickernell, C.J.W., Bard, J.A.M., **Drummond, D.A.**, "Stressful steps: progress and challenges in understanding stress-induced mRNA condensation and stress granule accumulation," *Molecular Cell* 82(14):2544–2556 (2022).
9. Yoo, H., **Drummond, D.A.**, "Using fluorescence anisotropy to monitor chaperone dispersal of RNA-binding protein condensates," *STAR Protocols* 3, 101409 (2022).
10. Yoo, H., Bard, J.B., Pilipenko, E.V., **Drummond, D.A.**, "Chaperones directly and efficiently disperse stress-triggered biomolecular condensates," *Molecular Cell* 82(4):741–755 (2022).
11. Cereghetti, G., Wilson-Zbinden, C., Kissling, V.M., Diether, M., Arm, A., Yoo, H., Piazza, I., Saad, S., Picotti, P., **Drummond, D.A.**, Sauer, U., Dechant, R., Peter, M., "Reversible amyloids of pyruvate kinase couple cell metabolism and stress granule disassembly," *Nature Cell Biology* 23:1085–1094 (2021).
12. Triandafillou, C.G., **Drummond, D.A.**, "Live cell measurement of the intracellular pH of yeast by flow cytometry using a genetically-encoded fluorescent reporter," *Bio-protocol* 10(12) :e3563 (2020).
13. Pattanayak G.K., Liao Y., Wallace E.W.J., Budnik B.A., **Drummond D.A.**, Rust M.J., "Daily cycles of reversible protein condensation in cyanobacteria," *Cell Reports* 32 (7):108032 (2020).
14. Triandafillou C.G., Katanski C.D., Dinner A.R., **Drummond D.A.**, "Transient intracellular acidification regulates the core transcriptional heat shock response," *eLife* 2020 (9):e54880 (2020).
15. Yoo, H., Triandafillou, C.G., and **Drummond, D.A.** "Cellular sensing by phase separation: Using the process, not just the products." *The Journal of Biological Chemistry* 294 (18): 7151–59 (2019).
16. McCully, A.L., Behringer, M.G., Gliessman, J.R., Pilipenko, E.V., Mazny, J.L., Lynch, M., **Drummond, D.A.**, McKinlay, J.B., "An *Escherichia coli* nitrogen starvation response is important for mutualistic coexistence with *Rhodopseudomonas palustris*," *Applied and Environmental Microbiology* e00404-18 (2018).
17. Riback, J.A., Katanski, C.D., Kear-Scott, J.L., Pilipenko, E.V., Rojek, A.E., Sosnick, T.R., and **Drummond, D.A.**, "Stress-triggered phase separation is an adaptive, evolutionarily tuned response," *Cell* 168(6):1028–1040 (2017).
18. Wallace, E. W. J., Kear-Scott, J. L., Pilipenko, E. V., Schwartz, M. H., Laskowski, P. R., Rojek, A. E., Katanski, C.D., Riback, J.A., Dion, M.F., Franks, A., Airoidi, E.M., Pan, T., Budnik, B.A., **Drummond, D. A.** (2015). "Reversible, specific, active aggregates of endogenous proteins assemble upon heat stress," *Cell* 162(6):1286–1298 (2015).

19. Csárdi, G., Franks, A.M., Choi, D.S., Airoidi, E.M., and **Drummond, D.A.**, “Accounting for experimental noise reveals that mRNA levels, amplified by post-transcriptional processes, largely determine steady-state protein levels in yeast,” *PLoS Genetics* 11(5): e1005206 (2015).
20. Zaborske, J., Bauer DuMont, V.L., Wallace, E.W.J., Pan, T., Aquadro, C.F., and **Drummond, D.A.**, “A nutrient-driven tRNA modification alters translational fidelity and genome-wide protein encoding across an animal genus,” *PLoS Biology* 12(12):e1002015 (2014).
21. Franks, A.M., Csárdi, G., **Drummond, D.A.**, Airoidi, E.M., “Estimating a structured covariance matrix from multi-lab measurements in high-throughput biology,” *Journal of the American Statistical Association*, DOI: 10.1080/01621459.2014.964404 (2014). (Winner of the 2015 ASA W. J. Youden Award.)
22. Wallace, E.W.J., Airoidi, E.M., and **Drummond, D.A.**, “Estimating selection on synonymous codon usage from noisy experimental data,” *Molecular Biology & Evolution* 30(6):1438–53 (2013). (Faculty of 1000 rated.)
23. Geiler-Samerotte, K.A., Hashimoto, T., Dion, M.F., Budnik, B.A., Airoidi, E.M., and **Drummond, D.A.**, “Quantifying condition-dependent intracellular protein levels enables high-precision fitness estimates,” *PLoS One*, 8(9): e75320 (2013).
24. Agashe, D., Martinez-Gomez, N.C., **Drummond, D.A.**, and Marx, C.J., “Good codons, bad transcript: large reductions in gene expression and fitness arising from synonymous mutations in a key enzyme,” *Molecular Biology & Evolution*, 30(3):549–60 (2013).
25. Geiler-Samerotte, K.A., Dion, M.F., Budnik, B.A., Hartl, D.L., and Drummond, D.A., “Misfolded proteins impose a dosage-dependent fitness cost and trigger a cytosolic unfolded protein response in yeast,” *Proc. Nat’l. Acad. Sci. USA* 180(2):680–685 (2011).
26. Wilke, C.O. and Drummond, D.A., “Signatures of protein biophysics in coding sequence evolution,” *Curr. Opin. Struct. Bio.* 20(3):385–9 (2010).
27. Drummond, D.A. and Wilke, C.O., “The evolutionary consequences of erroneous protein synthesis,” *Nature Reviews Genetics* 10(10):715–24 (2009).
28. Drummond, D.A. and Wilke, C.O., “Mistranslation-induced protein misfolding as a dominant constraint on coding-sequence evolution,” *Cell* 134(2):341–52 (2008). (Faculty of 1000 rated.)
29. Zhou, T., Drummond, D.A., and Wilke, C.O., “Contact density affects protein evolutionary rate from bacteria to mammals,” *J. Mol. Evol.* 66(4):395–404 (2008).
30. Li, Y., Drummond, D.A., Sawayama, A.M., Snow, C.S., Bloom, J.D., and Arnold, F.H., “A diverse family of thermostable cytochrome P450s created by recombination of stabilizing fragments,” *Nat. Biotech.* 25(9):1051–6 (2007). (Faculty of 1000 rated.)
31. Bloom, J.D., Drummond, D.A., Arnold, F.H., and Wilke, C.O., “Structural determinants of the rate of protein evolution in yeast,” *Molecular Biology & Evolution* 23(9):1751–61 (2006).
32. Wilke, C.O. & Drummond, D.A., “Population genetics of translational robustness,” *Genetics* 173:473–81 (2006).
33. Drummond, D.A., Raval, A., and Wilke, C.O., “A single determinant dominates the rate of yeast protein evolution,” *Molecular Biology & Evolution* 23(2):327–37 (2006). (MBE Early Citation Classic, 2013–2014; Faculty of 1000 rated.)
34. Drummond, D.A., Bloom, J.D., Adami, C., Wilke, C.O., and Arnold, F.H., “Why highly expressed proteins evolve slowly,” *Proc. Nat’l. Acad. Sci. USA* 102(40):14338–43 (2005). (Faculty of 1000 rated.)

35. Wilke, C.O., Bloom, J.D., Drummond, D.A., and Raval, A., "Predicting the tolerance of proteins to random amino acid substitution," *Biophysical Journal* 89(6):3714–20 (2005).
36. Drummond, D.A., Iverson, B.L., Georgiou, G.G., and Arnold, F.H., "Why high-error-rate mutagenesis libraries are enriched in functional and improved proteins," *J. Mol. Biol.* 350(4):806–16 (2005). (Faculty of 1000 rated.)
37. Drummond, D.A., Silberg, J.J., Meyer, M.M., Wilke, C.O., and Arnold, F.H., "On the conservative nature of intragenic recombination," *Proc. Nat'l. Acad. Sci. USA* 102(15):5380–5 (2005).
38. Bloom, J.D., Silberg, J.J., Wilke, C.O., Drummond, D.A., Adami, C., and Arnold, F.H., "Thermodynamic prediction of protein neutrality," *Proc. Nat'l. Acad. Sci. USA* 102(3):606–11 (2005). (Faculty of 1000 rated.)

(b) Non-peer-reviewed original articles

1. Triandafillou, C.G. and **Drummond, D.A.** "Heat Shock Factor 1: from fire chief to crowd-control specialist," *Mol. Cell* 63(1):1–2 (2016).
2. Wallace, E.W.J. and **Drummond, D.A.** "Dying mRNA tells a story of its life." *Cell* 161(6):1246–1248 (2015).
3. **Drummond, D.A.** "How infidelity creates a sticky situation." *Mol. Cell* 48:663–4 (2013).
4. **Drummond, D.A.** "Innovative chaps." *Current Biology* 19(17):R740-2 (2009).

(c) Other works that are publicly available (websites, interviews, publications in the popular press, testimony, computer programs, protocols, reagents, inventions, patents not listed above, etc.)

1. "Mysterious Blobs Found Inside Cells Are Rewriting the Story of How Life Works," *Scientific American* (cover article), interviewee, published 1/21/2025.
2. "Heat-inducible self-assembling protein domains." U.S. patent WO2016065273 A1, application number PCT/US2015/057139. Filed 10/23/2015, published 4/28/2016.

FUNDING

(a) Past:

1. NIH R01 GM088344. PI: Claus O. Wilke, University of Texas at Austin. My role: co-PI. Title: "The biophysical basis of translational selection." Project period: 2009–2013. Total direct costs: \$134,680 (for grant period at U.Chicago, 2011–2013). Annual salary recovery or effort: 17%.
2. Alfred P. Sloan Foundation Sloan Research Fellowship. PI: Drummond. My role: PI. Title: "Proteome-wide investigation of mistranslation-induced protein misfolding." Project period: 2012–2014. Total direct costs: \$50,000. Annual salary recovery or effort: 5%.
3. Big Ideas Generator (U.Chicago) Seed grant. PI: Drummond. My role: PI. Title: "Evolving adaptive protein assembly." Total direct costs: \$11,000. Annual effort: 2.5%, no salary recovery. Project period: 12/5/2014–6/5/2015.
4. Chicago Biomedical Consortium Postdoctoral Research Grant. PI: Drummond. My role: PI. Title: "Tracking mRNA localization and translation under heat stress." Total direct costs: \$15,000. Annual effort: 0%, no salary recovery. Project period: 7/1/2015–6/30/2016.
5. Pew Charitable Trusts, Pew Scholar in the Biomedical Sciences. PI: Drummond. My role: PI. Total direct costs: \$240,000. Project period: 8/1/2012–7/31/2016.

6. Big Ideas Generator (U.Chicago) travel grant. My role: PI. Title: "Visualizing quinary interactions." Total direct costs: \$5,000, travel only. Project period: 1/1/2016–9/1/2016.
7. ITM Collaborative Grant. My role: PI. Title: "Aggregation Pathology after Mild Traumatic Brain Injury." Total direct costs: \$25,000. Annual effort: 0%. Project period: 11/1/2015–10/30/2017.
8. NIH NIGMS 1-U54-GM105816. Lead PI: Patricia Clark, Notre Dame. **My role: PI.** Title: "Interactions regulating translation and protein biogenesis in vivo." Total direct costs (Drummond lab): \$148,719. Annual salary recovery or effort: 8.3%. Project period: 6/1/2013–4/30/2018.
9. US Army Research Office W911NF-14-1-0411. PI: Michael Lynch, Indiana University. **My role: co-PI.** Title: "Mechanisms of prokaryotic evolution." Total direct costs: \$701,870 (Drummond lab). Annual salary recovery or effort: 2% (1 week). Project period: 7/1/2014–6/30/2019.
10. Glenn Foundation FP071252-01-PR. PI: Drummond. **My role: PI.** Title: "Protein aggregation in aging." Total direct costs: \$60,000. Annual salary recovery or effort: none. Project period: 8/27/2018–8/26/2019.
11. NIH R01 GM126547. PI: Drummond. **My role: PI.** Co-PI: Sosnick. Title: "Determining the molecular basis of adaptive stress-triggered protein phase separation." Total direct costs: \$813,368. Annual salary recovery or effort: 30%. Project period: 12/1/2018–11/30/2022.
12. NIH R01 GM127406. PI: Drummond. **My role: PI.** Title: "Signatures of interaction-driven selection on low-complexity sequences." Total direct costs: \$700,000. Annual salary recovery or effort: 25%. Project period: 4/1/2018–3/31/2022.

(b) Current:

1. NIH R35 GM144278. PI: Drummond. **My role: PI.** Title: "Function and regulation of stress-induced adaptive condensates." Total direct costs: \$2,326,264. Annual salary recovery or effort: 60%. Project period: 1/1/2022–12/31/2026. Renewal anticipated, impact score 16, 1/1/2027–12/31/2031.
2. NIH RM1 GM153533. Contact PI: D. Pincus (MPI), PI: A. Squires, S. Oakes, A. Basu, M. Mani. **My role: Significant Collaborator.** Title: "Multimodal investigation of cellular adaptation across timescales." Annual salary recovery or effort: 0%. Total award amount: \$7,403,716. Project period: 8/1/24–7/31/29.

HONORS, PRIZES, AND AWARDS

1991	National Merit Scholarship
1995	<i>Cum laude</i> , Princeton University
2006	Graduate Student Poster Award, Society for Molecular Biology & Evolution annual meeting.
2006	Everhart Distinguished Graduate Student Lecturer Award
2006	Demetriades-Tsafka Prize in Bioengineering or Related Fields
2006	Milton and Francis Clauser Doctoral Prize
2008	Postdoctoral Fellow Poster Award, Society for Molecular Biology & Evolution annual meeting
2009	Symposium Fellow, 74th CSHL Symposium on Quantitative Biology (Evolution: The Molecular Landscape)
2012	Pritzker Scholar, The University of Chicago
2012	Ellison Medical Foundation New Scholar in Aging (declined)
2012–2014	Sloan Research Fellow
2012–2016	Pew Scholar in the Biomedical Sciences, Pew Charitable Trusts
2015	Best Poster prize, Biology of Molecular Chaperones conference (EMBO), Heraklion, Crete
2015	American Statistical Association W.J. Youden Award (with 3 coauthors)
2017	Professor Mariam Hasson Memorial Lecture, Purdue University
2017	Distinguished Investigator, The University of Chicago Biological Sciences Division
2022	Irvin Isenberg Memorial Lecturer, Marine Biological Laboratories
2022	Distinguished Blaffer Lecturer, MD Anderson Cancer Center
2023	Keynote speaker, 28 th Annual Midwest Stress and Molecular Chaperone Meeting

INVITED SPEAKING

2004	Research seminar, USC Biological Sciences, Los Angeles, CA.
2009	Invited lecturer, Frontiers in Genomics Seminar Series, UNAM, Cuernavaca, Mexico.
2010	Invited speaker, EMBO Conference Series, Experimental Approaches to Evolution and Ecology using Yeast, Heidelberg, Germany.
2011	Invited speaker, SMBE annual meeting, Kyoto, Japan 2011.
2011	Invited speaker, ASCB Symposium on Protein Balance, Denver, CO.
2012	Invited speaker, SMBE Phylomedicine Symposium, Tempe, AZ.
2012	Research seminar, Computations in Science seminar series, University of Chicago, Chicago, IL.
2012	Research seminar, Chicago Mass Spectrometry Discussion Group, Northwestern University, Chicago, IL.
2012	Research seminar, University of Rochester, Rochester, NY.
2012	Closing plenary speaker, Midwest Conference on Protein Folding.
2012	Research seminar, KU Leuven, Belgium.
2012	Invited speaker, VIB Student Symposium, Ghent, Belgium.
2012	Research seminar, University of Wyoming, Laramie, WY.
2012	Research seminar, Stony Brook University, Stony Brook, NY.
2013	Research seminar, Kavli Frontiers of Science Symposium, Irvine, CA.
2013	Research seminar, Princeton University, Princeton, NJ.
2014	Invited speaker, SMBE 'Causes of Genome Evolution' Symposium, Mishima, Japan.
2014	Invited speaker, 1st HSF workshop, Paris, France.
2014	Research seminar, Columbia University, New York, NY.

2014 Research seminar, University of Texas Medical School, Houston, TX.
 2015 Research seminar, University of New Hampshire, Durham, NH.
 2015 Research seminar, Fred Hutchinson Cancer Research Center, Seattle, WA.
 2015 Invited speaker, KITP Evolutionary Cell Biology, Summer Research Course.
 2015 Invited speaker, Midwest Yeast Meeting, Chicago, IL.
 2015 Invited speaker, Gordon Research Conference on Stress Proteins in Growth, Development and Disease, Lucca, Italy.
 2015 Research seminar, Scripps Research Institute, La Jolla, CA.
 2016 Research seminar, Stanford University, Palo Alto, CA.
 2016 Research seminar, Marquette University, Milwaukee, WI.
 2016 Research seminar, Max Planck Institute of Molecular Cell Biology and Genetics, Dresden, Germany.
 2016 Invited speaker, Physical Basis of Cellular Adaptation and Memory, Barbados.
 2016 Research seminar, Case Western Reserve University, Cleveland, OH.
 2016 Invited speaker, Protein Folding in the Cell, FASEB Science Research Conference, Saxton's River, VT.
 2016 Research seminar, University of Utah, Salt Lake City, UT.
 2016 Research seminar, Johns Hopkins University (Biology), Baltimore, MD.
 2016 Research seminar, University of Michigan, Ann Arbor, MI.
 2016 Research seminar, Northwestern University, Chicago, IL.
 2016 Research seminar, University of Colorado School of Medicine, Denver, CO (declined).
 2016 Research seminar, ETH Zurich, Zurich, Switzerland.
 2016 Invited speaker, Joint Workshop on Protein Biology and Technology, University of Copenhagen, Copenhagen, Denmark (declined).
 2016 Invited speaker, University of Pennsylvania, Philadelphia, PA.
 2017 Invited speaker, Biophysical Society, Biopolymers In Vivo subgroup, New Orleans, LA.
 2017 Invited speaker, Université de Montreal, Montreal, Quebec, Canada.
 2017 Invited speaker, Bellairs Workshop, The Physical Basis of Cellular Adaptation and Memory, Barbados.
 2017 Invited speaker, Physiological Genomics, Experimental Biology 2017, Chicago, IL.
 2017 Research seminar, UCSF (Biochemistry & Biophysics), San Francisco, CA.
 2017 Research seminar, Stowers Institute for Medical Research, Kansas City, MO.
 2017 Research seminar (Professor Mariam Hasson Memorial Lecture), Purdue University, West Lafayette, IN.
 2017 Invited speaker, Protein Aggregation in Health & Disease, FASEB Science Research Conference, Steamboat Springs, CO.
 2017 Invited speaker, Gordon Research Conference on Stress Proteins in Growth, Development, and Disease, Sunday River, ME.
 2017 Research seminar, UT Southwestern, Dallas, TX.
 2017 Invited speaker, University of Toronto, Dept. of Molecular Genetics Colloquium Series, Toronto, Canada.
 2017 Invited speaker, 15th Chinese Biophysics Congress, Shanghai, China.
 2018 Invited speaker, Biophysical Society, Biopolymers In Vivo subgroup annual meeting.
 2018 Invited speaker, EMBO|EMBL Cellular Mechanisms Driven by Liquid Phase Separation, Heidelberg, Germany.
 2018 Invited fellow, Scialog fellows meeting.

2018 Invited speaker, Networks of Cellular Surveillance Mechanisms, Heidelberg, Germany.

2019 Invited speaker, RIKEN Institute, Tokyo, Japan.

2019 Invited speaker, Federation of the Asian and Oceanian Physiological Societies, Kobe, Japan.

2019 Invited speaker, Keystone Symposium on Biomolecular Condensates, Snowbird, UT.

2019 Invited speaker, Genetics Training Program, University of Michigan.

2020 Invited speaker, IDP Seminars (online).

2020 Invited speaker, Cell and Protein Homeostasis webinars (online).

2021 Invited speaker, FASEB Catalyst Conference on Extracellular and Organismal Proteostasis in Health and Disease (online).

2021 Invited session keynote and session chair, Aggregates and Intrinsically Disordered Proteins II, Biophysical Society annual meeting (online).

2021 Invited speaker, UC Berkeley MCB seminar series (online).

2021 Invited speaker, Constrained & Directional Evolution meeting, Japan (online).

2021 Invited speaker, UToronto RNA Club (online).

2021 Session chair, FASEB SRC conference on Protein Aggregation (online).

2021 Invited speaker, Plasticity in Biological Organization conference, Telluride, CO.

2021 Invited speaker, Proteostasis Consortium (online).

2021 Invited speaker, *Cell* Symposium: Biological Assemblies: Phase Transitions and More (online).

2022 Invited speaker, Emory University, Atlanta, GA.

2022 Invited speaker, Condensate Colloquium (online).

2022 Invited speaker, Gordon Research Conference on Intrinsically Disordered Proteins, Les Diablerets, Switzerland.

2022 Irvin Isenberg Memorial Lecturer, MBL Physiology Course, Woods Hole, MA (endowed lectureship).

2022 Distinguished Blaffer Lecturer, MD Anderson Cancer Center, Houston, TX.

2022 Invited speaker, Kitchen Table Talks, Condensates.com/Dewpoint (online).

2023 Keynote speaker, 28th Annual Midwest Stress and Molecular Chaperone Meeting, Chicago, IL.

2023 Session keynote, "Cellular Matters: toward an understanding of the properties, functions & applications of bio-condensates" (BML/ETH Zurich), Ascona, Switzerland.

2023 Invited speaker, Plasticity in Biological Organization, Telluride, CO.

2023 Invited speaker, Susan Lindquist School on Proteostasis, Frankfurt, Germany.

2023 Invited speaker, RNA Granules 2023, Surrey, UK.

2023 Invited speaker, Biochemistry, Biophysics & Biodesign Seminar Series, CCNY and CUNY Advanced Science Research Center, New York, NY (online).

2024 Invited speaker, Phase Separation Interest Group, UCSF/NIH (online).

2024 Invited speaker, University of Wisconsin, Madison, WI.

2024 Invited speaker, University of Pittsburgh, Pittsburgh, PA.

2024 Invited speaker, Princeton University, Princeton, NJ.

2024 Invited speaker, Johns Hopkins University, Baltimore, MD.

2024 Invited speaker, RNA Regulation, Stowers Institute, Kansas City, MO.

2025 Invited speaker, Vanderbilt University, Nashville, TN.

2025 Impact Lecture (student-invited), Biochemistry, Notre Dame, South Bend, IN.

2025 Invited speaker, Pizza with the PIs, UChicago.

2025 Invited speaker, ASBMB annual meeting, "Biochemistry and climate change: Molecular responses and innovations," Chicago, IL.

2025	Invited speaker, Biological of Biomolecular Condensates: Bridging Theory and Experiment, Kavli Institute for Theoretical Physics, Santa Barbara, CA.
2025	Invited speaker, EMBO/FEBS Lecture Course: Susan Lindquist School on Proteostasis, Espoo, Finland.
2025	Invited speaker, Physical Properties of the Cytoplasm (online).
2026	Invited speaker, Genetics seminar series, UW-Madison, Madison, WI.
2026	Invited speaker, Polyelectrolytes in Biology, Berlin, Germany.
2026	Invited speaker, EMBL Cellular Mechanisms Driven by Phase Separation, Heidelberg, Germany.
2026	Invited speaker, Post-Transcriptional Gene Regulation Gordon Research Conference, Portland, ME.
2027	Invited speaker, American Society for Biochemistry & Molecular Biology annual meeting, Boston, MA.

INVITED, ELECTED, OR APPOINTED EXTRAMURAL SERVICE

Various	Guest editor: PNAS; PLOS Genetics; PLOS Computational Biology
Various	Reviewer: Science; Nature; Cell; PNAS; eLife; PLOS Biology; PLOS Genetics; Molecular Cell, Molecular Biology and Evolution
2017	Co-organizer, Biopolymers In Vivo subgroup symposium 2018, Biophysical Society (elected)
2017	Co-organizer, Protein Society 2018 annual meeting (invited)
2017	Faculty mentor, Gordon Research Seminar on Stress Proteins in Growth, Development, and Disease, Sunday River, ME (invited)
2020	Faculty instructor, Leading the Lab: Effective Lab Management, Society for Developmental Biology (virtual, invited)
2021	Poster reviewer, Midwest Stress Response and Molecular Chaperones meeting (virtual, invited)
2021	Session chair, "Platform: Intrinsically Disordered Proteins (IDP) and Aggregates II", Biophysical Society 65 th Annual Meeting (virtual, invited)
2021	Session chair, Protein Aggregation, FASEB meeting, "The Protein Aggregation Conference: Function, Dysfunction, and Disease" (virtual, invited)
2021	Panelist, Visualizing Protein Age, The Paul G. Allen Frontiers Group (virtual, invited)
2022	Panelist, Gordon Research Seminar, Intrinsically Disordered Proteins, Les Diablerets, Switzerland (invited)
2022	NIH R35 (MIRA) review panelist (virtual, invited)
2022	NSF CAREER reviewer (invited)
2025	NIH CSF1 study section (invited)
2025	Organizer, Condensate Day 2.0, National Institute for Theory and Mathematics in Biology (invited)
2026	NIH ZRG1 study section (invited)

PROFESSIONAL SOCIETIES

Elected or invited membership:

None

Other:

American Association for the Advancement of Science
American Society for Cell Biology
American Society for Biochemistry & Molecular Biology

Biophysical Society

EDUCATION

The College (B.A., B.S.):

- 2013–2019 BCMB/HGEN/ECEV 31100, “Evolution of Biological Molecules.” Winter Quarter, 2 lecturers, open to undergraduates with instructor permission (one undergraduate student in winter 2019)
- 2011– Undergraduate research mentor, eight students
- 2020– Biology Honors. 16-25 students/year. Course director and instructor. 2020/2021, member of all 16 thesis committees. 50 contact hours.

Graduate programs (Ph.D.):

- 2011–2016 Yearly lecture on mass spectrometry, BCMB 304, “Protein Fundamentals”
- 2013–2019 BCMB/HGEN/ECEV 31100, Evolution of Biological Molecules. Winter quarter, co-taught with Joe Thornton (ECEV/HGEN), 21 students in 2017. Half of course or ten 2-hour lectures (class time extended to 2 contact hours per session).
- 2015–2017 GENE 31800, Current Topics in Genetics. Winter quarter. Required ½-credit course for all GGSB students, centered on the student-invited seminar series. 5 contact hours.
- 2016– BPHYS 40500, Research Presentations. Spring. Course creator and instructor. 15 contact hours.
- 2023– MGCB 33500, Fundamentals of Biological Data Analysis (Rust/Munro/Drummond). 6 lectures.

Other:

- 2013–2021 Creator and coordinator, Intro to Programming and intermediate programming workshops. BSD-wide, 42+ students. 2016: one-day intro workshop, two-day intermediate workshop. 2 contact hours.

Research trainees:

- (a) High school students and teachers
None.

(b) Undergraduate (B.A., B.S.)

- 2011–2012 Allison Weaver, The University of Chicago
- 2011–2013 Joseph Park, The University of Chicago
- 2012–2013 Klevin Lo, The University of Chicago
- 2012–2015 Alexandra Rojek, Harvard University. Drummond was sole mentor for Rojek’s Honors Thesis research. Rojek’s thesis was awarded the Hoopes Prize (thesis prize, all fields, ~50 winners/year) and the Laurence J. Henderson Prize (biochemical sciences thesis prize, 1 winner/year).
- 2013–2015 Eliza Nieweglowska, The University of Chicago (Honors Thesis mentor)
- 2014–2016 Rohan Goyal, The University of Chicago
- 2015–2017 Daniel Ruttenberg, The University of Chicago
- 2017–2019 Aparna Srinivasan, The University of Chicago
- 2018–2022 Wenshan (Rosalind) Pan, University of Chicago
- 2020–2021 Honors thesis committee member for Christopher Dann
- 2020–2021 Honors thesis committee member for Esther Ochoa
- 2020–2021 Honors thesis committee member for Harrison Greenbaum
- 2020–2021 Honors thesis committee member for Jonathan Zhou

2020–2021 Honors thesis committee member for Bonnie Hu
 2020–2021 Honors thesis committee member for Maymay Ou
 2020–2021 Honors thesis committee member for Eric Chen
 2020–2021 Honors thesis committee member for Katie DeLong
 2020–2021 Honors thesis committee member for Grace Ryan
 2020–2021 Honors thesis committee member for Ruby Schnirman
 2020–2021 Honors thesis committee member for Arielle Weinstein
 2020–2021 Honors thesis committee member for Hubert Myslewicz
 2020–2021 Honors thesis committee member for Lia Jueng
 2020–2021 Honors thesis committee member for Maya Waarts
 2020–2021 Honors thesis committee member for Ali Veinbachs
 2020–2021 Honors thesis committee member for Shane Warland
 2021–2022 Honors thesis committee member for Matthew Sabados
 2021–2022 Honors thesis committee member for Rosalind Pan
 2022–2025 Dana Christopher, The University of Chicago
 2022–2025 Josh Melamed, The University of Chicago (Honors)
 2023–2024 Honors thesis committee member for Manoj Bhangwat
 2023– Jeffery Zhang, The University of Chicago
 2024–2025 Thesis committee member for Diangen (Dana) Lin
 2025– Madison Nelson, The University of Chicago
 2026– Jiawei Ge, The University of Chicago

(c) Medical (M.D.)

None

(d) Graduate (Ph.D.)

Harvard University

2006–2011 **Primary supervisor** for Kerry Geiler-Samerotte (co-supervised by Daniel Hartl)
 2008 Dissertation examiner for Trevor Bedford

The University of Chicago

2013–2018 **Co-supervisor** for Joshua Riback, BPHYS (required dual mentorship; Tobin Sosnick was co-supervisor). Riback received a Harper Fellowship for 2017–2018. Ph.D. granted 2018.
 2012–2017 Ph.D. committee member and chair for Wesley Clark, BCMB.
 2012–2019 Ph.D. committee member for Katherine Proudfoot, MGCB.
 2013–2018 Ph.D. committee chair for Tyler Starr, BCMB.
 2014–2018 Ph.D. committee member for Aarti Venkat, Human Genetics.
 2014–2019 Ph.D. committee member for Mohammed Siddiq, Ecology & Evolution.
 2013–2018 Ph.D. committee member for Michael Schwartz, Microbiology.
 2014–2018 Ph.D. committee member for Katherine Zhou, BCMB.
 2014–2019 **Primary supervisor** for Christopher Katanski, BCMB. Ph.D. granted 2019.
 2014–2020 **Co-supervisor** for Catherine Triandafillou, BPHYS (required dual mentorship; Aaron Dinner is co-supervisor). Ph.D. granted 2020.
 2015–2021 Ph.D. committee member for Arvind Pillai, Ecology & Evolution.
 2016–2021 Ph.D. committee member for Matthew Reyer, BPHYS.
 2016–2022 Ph.D. committee member for Charles Lang, GGSB.
 2016–2021 **Primary supervisor** for Haneul Yoo, BCMB. Ph.D. granted 2021. *Yoo received the 2022 Best Overall Dissertation Award for the Biological Sciences Division.*
 2017–2020 External Ph.D. committee member for Nicole Nuckolls, Stowers Institute.

2018–2022	Ph.D. committee chair for Yeonwoo Park, ECEV. Ph.D. 2022.
2018–2023	Primary supervisor for Samantha Keyport Kik, GGSB. Ph.D. granted 2023.
2018–2023	Co-supervisor for Hendrik Glauninger, BPHYS/MSTP (required dual mentorship; Tobin Sosnick is co-supervisor). Ph.D. granted 2023.
2019–2023	Ph.D. committee member for Philipp Ross, GGSB/Immunology. Ph.D. 2023.
2019–2025	Ph.D. committee member for Lily Burton, BCMB.
2019–2025	Ph.D. committee chair for Craig DeValk, BCMB.
2019–2024	Primary supervisor for Caitlin Wong Hickernell, BCMB. Ph.D. granted 2024.
2021–2025	Co-supervisor for Kyle Lin, BPHYS/MSTP (required dual mentorship; Allison Squires [PME] is co-supervisor).
2021–	Ph.D. committee member for Luke Dyer.
2021–	Ph.D. committee member for Jelena Scepanovic, Human Genetics.
2023–	Ph.D. committee chair for Caroline Kaminsky, BCMB.
2023–	Co-supervisor for Leah Chaney, CMB (David Pincus is co-supervisor).
2023–	Co-supervisor for Estefania Cuevas-Zepeda, BPHYS (required dual mentorship; Tobin Sosnick is co-supervisor).
2023–	Primary supervisor for Karen Velez, CMB.
2023–	Ph.D. committee chair for Maya Igarashi, BPHYS.
2024–	Thesis committee member for Dylan Stermer, GGSB.
2024–	Primary supervisor for Alyssa Guzman, BCMB.
2025–	Thesis committee member for Jermaine Austin, CCB.
2025–	Thesis committee member for Serena Debesai, BPHYS.

Various	Lab rotation supervisor for Wesley Clark (BCMB), Tyler Starr (BCMB), Andrei Anghel (CMB), Matthew Tien (BCMB), Siyi Gu (BCMB), Kevin Hunt (BCMB), Catherine Triandafillou (BPHYS), Catherine Wu (GGSB), Haneul Yoo (BCMB), Soo Ji Kim (BCMB), Julia Shangguan (BCMB, 2021), Pinky Lee (DRSB, 2021), Caroline Kaminsky (BCMB, 2022), Michelle Chiu (BPHYS, 2022), Arianna Stancari (BCMB, 2023), Brendan Jamison (GGSB, 2023), Leah Chaney (CMB, 2023), Karen Velez (CMB, 2023), Estefania Cuevas-Zepeda (BPHYS, 2023), Alyssa Guzman (BMB, 2023), Eric Leisten (Cancer Biology, 2024), Luna Han (BCMB, 2024), Lily Leveque-Eichhorn (CMB, 2024), Nadav Goldfajn (BPHYS, 2025), Sean Martin (CMB, 2026), Annie Corbett (BMB, 2026).
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(e) Postdoctoral

2010–2015	Edward W.J. Wallace, Ph.D.
2018–2023	Jared A.M. Bard, Ph.D.
2020–2021	Catherine Triandafillou, Ph.D. (post-graduation research)
2021–2022	Haneul Yoo, Ph.D. (post-graduation research)
2023	Samantha Keyport Kik, Ph.D. (post-graduation research)
2024–2025	Caitlin Wong Hickernell, Ph.D. (post-graduation research)

(f) Other

2012–2013	Advisor for Andrzej Bylicki, Polish master's student
2013–2014	Advisor for Pawel Laskowski, Polish master's student
2014–2015	Advisor for Jakub Kucharczyk, Polish master's student
2014–2015	Ph.D. committee member and examiner for Alexander Franks, Dept. of Statistics, Harvard University (advisor: Prof. Edo Airoldi)
2015–2016	Advisor for Weronika Danecka, Polish master's student
2016	Advisor for Feng Xing, Tsinghua University REU student

2017 Advisor for Yaoyi Chen, Tsinghua University REU student

CLINICAL

None

SERVICE

University of Chicago

Committee membership:

2011–2015	Graduate Recruiting Committee, Molecular Biosciences
2013–2016	Graduate Nominations Committee, Biological Sciences Division
2012–2013	Decorations Committee, Biochemistry & Molecular Biology
2014–2018	Research Reputation Committee, Biological Sciences Division
2015–2020	Graduate Recruiting Committee, Graduate Program in Biophysics
2019–2022	Graduate Recruiting Committee, Graduate Program in Biochemistry & Molecular Biophysics (chair)
2022–2023	Graduate Recruiting Committee, Graduate Program in Biochemistry & Molecular Biophysics (co-chair)
2011–	Curriculum Committee, Biochemistry & Molecular Biology
2014–2024	Steering Committee, Committee on Genetics, Genomics & Systems Biology
2018–	Advisory Committee on Assistant Professors, Biological Sciences Division
2020–	Diversity Committee, Biochemistry & Molecular Biology
2021–2024	Faculty Advisory Committee, Biological Sciences Division (elected)
2021–	Governing Committee, Biological Sciences Collegiate Division
2022–2024	Student Affairs Committee, Committee on Genetics, Genomics & Systems Biology
2025–2026	BMB Chair Search Committee

Other:

2011–	Interviewer of MSTP candidates, Pritzker School of Medicine
2012	Preliminary examiner for Matthew Hope and Molly Evans, BCMB
2012	Panelist, “From Academia to Industry and Back,” UChicago postdoctoral association panel
2012–	Host for visiting speakers for seminar series (Alice Ting, MIT; Christine Dunham, Emory; Edward Marcotte, UT Austin; Nobuhiko Tokuriki, UBC; Michelle Meyer, BC; Judith Frydman, Stanford; James Fraser, UCSF; Tanja Mittag, St. Jude); Anthony Hyman, MPI-CBG (Biophysics series)
2013	Preliminary examiner for Alicija Janik, BCMB
2013	Reviewer for Ellison Medical Foundation internal competition applications
2014	Recruiter/ambassador for U.Chicago at University of Maryland Baltimore County Meyerhoff Scholars program and McNair Scholars program presentations (underrepresented minority, high-achieving students)
2014	Evening research keynote for first Discover UChicago recruiting event (underrepresented minority recruiting)
2015	Guest presenter, UCMC Trustees development subcommittee
2015–2017	Faculty Advisory Board member, Big Ideas Generator (BIG)
2015	Guest speaker, annual UCMC Development retreat
2015	Preliminary examiner for Adam Zmyslowski
2015	Panelist, myCHOICE Academic Research Career Panel
2015	Invited seminar speaker, myCHOICE: “Leading and managing an academic research group”

2016	Panelist, "From Academia to Industry and Back," UChicago BSD Postdoctoral Association panel
2017	Panelist, The Women's Board meeting, "Discovery Science & Human Health: What's at Stake?"
2017	Faculty host, BSD Postdoctoral Association Mentoring Roundtable, "Starting and Managing a New Lab"
2017	Organizer and host, GGSB 2017 Annual Symposium: "A new frontier in cellular structure and function"
2018	Preliminary examiner for Yifei Hu, BCMB
2018	Preliminary examiner for Craig DeValk, BCMB
2019	Preliminary examiner for Abigail Schroeter, BCMB
2019	Preliminary examiner for Jaeda Patton, GGSB
2019	Preliminary examiner for Szymon Kordon, BCMB
2020	Preliminary examiner for Sebastien Bastide, GGSB
2020	Preliminary examiner for Yunqi Yang, GGSB
2020	Preliminary examiner for Urszula Cichon, BCMB
2021	Preliminary examiner for Olivia Pura, BCMB
2021	Preliminary examiner for Daniel Krochmal, BCMB
2022	Faculty candidate host for Andrea Putnam and Steven Boeynaems (BMB)
2022–2025	Review committee member, IBD Scholars
2022	Ad hoc faculty review committee, BSCD Mastership
2022–	Panelist, Student Research Society
2022–	Faculty advisor, Student Research Society (registered SRO)
2025	Review committee member, internal Pew Biomedical Scholars competition
2025	Preliminary examiner for Camryn Shmelzer, CMB
2025	Preliminary examiner for Yusuf Noffel, ISTP
2026	Preliminary examiner for Tanvi Brevig, BCMB
2026	Preliminary examiner for Jessie Bolger, BCMB

Extramural (not indicated above)

Leadership:

2014	Session chair, "Protein folding on the ribosome" session. Midwest Stress and Cellular Chaperones conference, Northwestern University, Chicago, IL.
2016	Co-organizer, Midwest Stress and Cellular Chaperones conference, Northwestern University, Chicago, IL.
2017	Faculty mentor, Gordon Research Symposium, Stress Proteins in Health & Disease, Sunday River, ME.
2017	Program co-organizer, Biopolymers in Vivo subgroup, Biophysical Society annual meeting, San Francisco 2018.
2017	Co-organizer, Protein Society annual meeting, Boston 2018.
2021	Session chair, "The Protein Aggregation Conference: Function, Dysfunction, and Disease," FASEB (virtual).

Other:

2013	Panelist, "Publishing High Impact," sponsored by postdoctoral association, Northwestern University, Chicago, IL.
2023	Speaker, "Leading the Lab," Society for Developmental Biology (YouTube).
2024–	Activity booth host and presenter, South Side Science Festival, Chicago, IL.